

PES University — Dept. of CSE(AIML)

LAB 1: REQUIREMENTS ENGINEERING & UML USE-CASE MODELLING

Problem Statement #35 — Customizable Subscription Box Scheduler

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1. Problem Context

A subscription box portal where subscribers customize monthly product selections (such as books, coffee, and snacks) based on preference tags, with support for pausing or skipping billing cycles.

2. Target Stakeholders / Actors

Actor	Role in the System
Subscriber	Customizes monthly box selections, pauses or skips billing cycles, views order and billing history, and applies promo codes at checkout.
Fulfillment Lead	Generates the fulfillment manifest and manages shipping and inventory operations for the current billing cycle.
Payment / Billing Service	Processes or confirms billing transactions for subscription renewals and promo-code-adjusted charges.

3. Functional Requirements

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow subscribers to swap monthly box items or pause deliveries up to 48 hours prior to the monthly billing renewal date.	High	A change made before the 48-hour cut-off is saved and the updated selection is displayed.	Core customization requirement.
FR-002	Functional	The system shall allow subscribers to pause or skip an eligible billing cycle.	High	Selecting Pause or Skip changes the cycle status and displays the updated status.	Supports pause/skip capability.
FR-003	Functional	The system shall display the subscriber's	Medium	Opening the monthly box	Helps informed

		current box selection and preference tags before customization.		displays current items and preference tags.	customization.
FR-004	Functional	The system shall allow subscribers to apply a valid promo code during checkout and show the resulting charge.	Medium	A valid code displays the discount and correct final charge.	Supports promo-code checkout.
FR-005	Functional	The system shall allow the Fulfillment Lead to generate the fulfillment manifest for the current billing cycle.	High	The generated manifest contains all eligible boxes and selected items.	Required for fulfillment.

4. Non-Functional Requirements

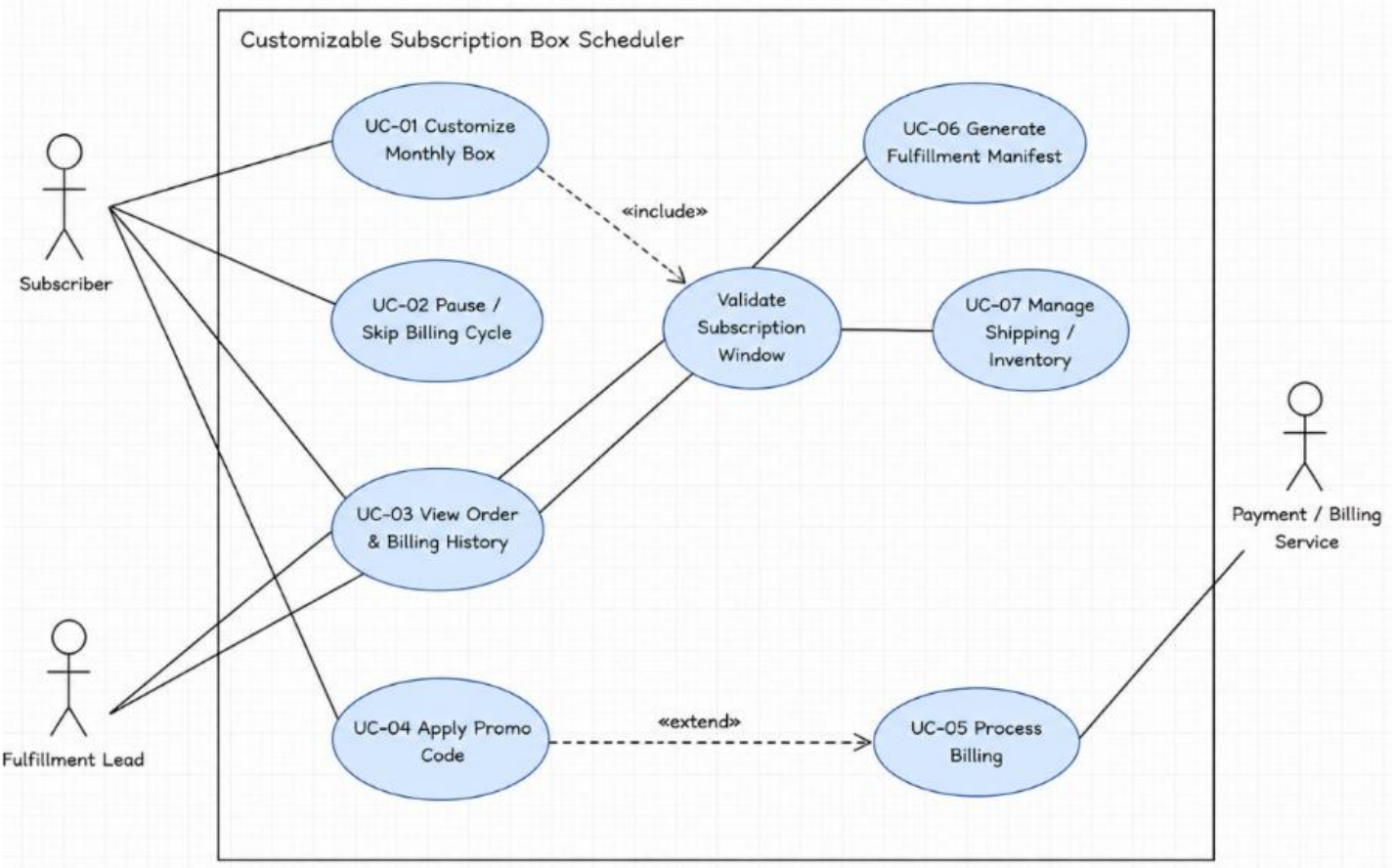
Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance	The fulfillment manifest generator shall export shipping labels for 10,000 monthly boxes in under 1 minute.	High	A benchmark with 10,000 boxes completes the export in less than 60 seconds.	Ensures acceptable peak-load performance.
NFR-002	Security	The system shall prevent unauthorized users from accessing subscriber billing and order information.	High	An unauthorized user is denied access and protected data is not displayed.	Protects subscriber information.
NFR-003	Availability	The subscription portal shall be available at least 99.5% of the time during each calendar month.	Medium	Monthly monitoring shows uptime of 99.5% or higher.	Supports reliable subscription management.
NFR-004	Usability	The system shall display confirmation or error feedback within 2 seconds after a subscriber submits	Medium	In usability testing, feedback appears within	Provides clear and responsive interaction.

		a customization request.		2 seconds for at least 95% of requests.	
NFR-005	Data Integrity	The system shall preserve the latest valid monthly box selection without loss when a subscriber completes a successful update.	High	After a successful update, reopening the subscription shows exactly the saved selection.	Prevents incorrect fulfillment caused by lost updates.

5. Use-Case Catalogue

ID	Use Case	Primary Actor	Related Requirements
UC-01	Customize Monthly Box	Subscriber	FR-001, FR-003
UC-02	Pause / Skip Billing Cycle	Subscriber	FR-002
UC-03	View Order & Billing History	Subscriber	NFR-002
UC-04	Apply Promo Code	Subscriber	FR-004
UC-05	Process Billing	Payment / Billing Service	FR-004
UC-06	Generate Fulfillment Manifest	Fulfillment Lead	FR-005, NFR-001
UC-07	Manage Shipping / Inventory	Fulfillment Lead	FR-005

6. UML Use-Case Diagram



7. UML Relationship Summary

Relationship	Description
UC-01 «include» Validate Subscription Window	Customization always checks whether the subscriber is within the allowed 48-hour window.
UC-04 «extend» UC-05 Process Billing	Applying a promo code is optional; billing can proceed without a promo code.

8. Use-Case Flow Specification — UC-01 Customize Monthly Box

Item	Specification
Primary Actor	Subscriber
Trigger	Subscriber opens the current monthly box to modify it.
Preconditions	Subscriber has an active subscription; the current monthly box is available; the customization window is open.
Postconditions	Updated box selection is saved and available to fulfillment; subscriber receives confirmation.

9. Main Success Scenario

1. The Subscriber opens the current monthly subscription box to view and modify the items selected for the upcoming billing cycle.
2. The System displays the subscriber's current box items along with the relevant preference tags to help the subscriber make suitable selections.
3. The Subscriber selects a replacement item or modifies the existing box selection according to their preferences.
4. The System verifies that the subscriber has an active subscription and that the requested modification is made before the 48-hour customization cut-off.
5. The System checks whether the selected replacement item is currently available for the subscription box.
6. If the selected item is available and all validations are successful, the System saves the updated monthly box selection.
7. The System displays a confirmation message informing the Subscriber that the monthly box has been successfully updated.
8. The System makes the updated box selection available to the Fulfillment Lead for inclusion in the fulfillment manifest for the current billing cycle.

10. Alternate Flow — Customization Window Closed

4A. The System determines that the Subscriber's request was made within the final 48 hours before the monthly billing renewal.

4A1. The System prevents the Subscriber from modifying the current monthly box selection because the customization period has ended.

4A2. The System displays a clear message informing the Subscriber that customization is locked for the current billing cycle.

4A3. The System keeps the original box selection unchanged and terminates the customization process.